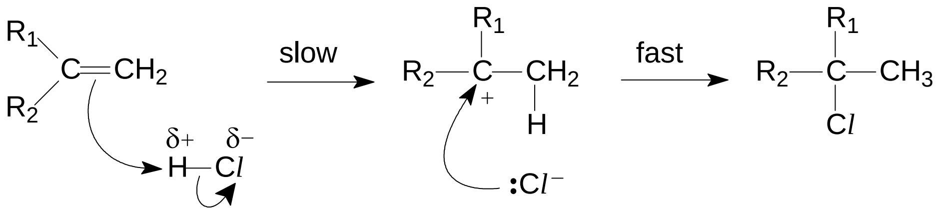
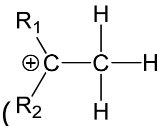
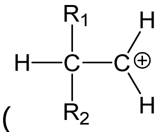
Section A

1 (a) (i) Electrophilic addition [3]



[1]: type of reaction
[1]: structure of carbocation intermediate, correct label of slow/fast
[1]: correct dipoles, lone pairs and curly arrows

(ii) [1]

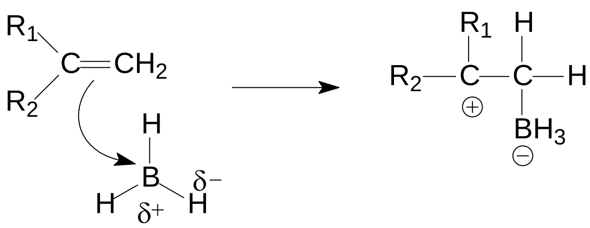
In step 1, the more stable (tertiary) carbocation intermediate () is formed instead of a (primary) carbocation ().

Alkyl groups exert an electron–donating effect, helping to disperse the positive charge on the carbocation, stabilising it. [1]

[1]: need to show clearly which carbocation is more stable – in drawing or words
[1]: correct explanation (do not accept H atom attaches itself to the double–bonded C atom already carrying a larger number of H atoms)

(b) (i) There are only 6 electrons around the central B atom of borane (incomplete octet) / there is an empty (p) orbital on boron. [1]

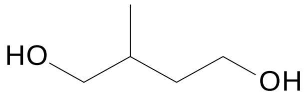
(ii) [2]

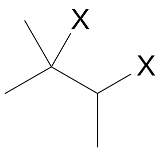


Not required
"favours formation of anti–Markovnikov product." and the structure of the intermediate showing presence of $–BH_3^-$
⇒ B atom is the electrophilic centre.

[1]: electrophilic B atom attaches itself to the double–bonded C carrying fewer both H atoms
[1]: correct dipoles on B–H bond and curly arrow

(iii) O.N. of C bonded to $–BH_2$ changes from –3 to –1. [1]
H_2O_2 acts as an oxidising agent. (awarded only if 1st mark is correct) [1]

(iv)  [1]

(c) **(i)**  [1]

X = C*l*, Br, I, OH

(ii) Step I: suitable halogen <u>X_2</u>, in inert solvent / <u>cold $KMnO_4$</u>, dil. H_2SO_4 [1]
chosen halogen should correspond with proposed structure of **H** provided in **(c)(i)**

Step II: <u>NaOH in ethanol, heat (under reflux)</u> / <u>excess conc. H_2SO_4, 180 °C</u> [1]

(iii) <u>Heat</u> separate samples of **G** and isoprene with <u>acidified $KMnO_4$</u>. [1]

G decolorises purple $KMnO_4$ <u>without effervescence</u>. [1]
Isoprene decolorises purple $KMnO_4$ along <u>with effervescence</u>.

(d) **(i)** $\mathbf{P_4O_{10}(s) + 6H_2O(l) \longrightarrow 4H_3PO_4(aq)}$ [1]

ignore wrong state symbols

(ii) For Al_2O_3, aluminum oxide is <u>insoluble</u> in water. <u>pH = 7</u>. [1]

For MgO, magnesium oxide <u>dissolves sparingly</u> in water to give a <u>white suspension of $Mg(OH)_2$</u>. <u>pH = 9</u>. [1]

$MgO(s) + H_2O(l) \rightleftharpoons Mg(OH)_2(s)$ [1]
(accept $\longrightarrow$)

2 (a) (i) *Brønsted–Lowry acid* is a proton (H^+) donor. [1]

(ii) $H_2SO_4 \rightleftharpoons HSO_4^- + H^+$ [1]
$HSO_4^- \rightleftharpoons SO_4^{2-} + H^+$

(accept $\longrightarrow$ for 1st step)

(iii) $[H^+]$ = **10 mol dm^{-3}** [1]
$[HSO_4^-]$ = **10 mol dm^{-3}**

(iv) [1]

$$K_a = \frac{[H^+][SO_4^{2-}]}{[HSO_4^-]}$$

(v) Let the $[H^+]$ from 2nd stage be x.

$$\frac{(10+x)(x)}{(10-x)} = 1.0 \times 10^{-2}$$

Since x << 10, (10 + x) ≈ 10 and (10 – x) ≈ 10
x = **0.0100 mol dm^{-3}** [1]

$[H^+]_{total}$ = 10 + 0.0100
= **10.01 mol dm^{-3}** [1]

Alternative

$$\frac{(10+x)(x)}{(10-x)} = 1.0 \times 10^{-2}$$

$10x + x^2 = 0.1 - 0.01x$
$x^2 + 10.01x - 0.1 = 0$
solving by GC, x = **0.00998**

$[H^+]_{total}$ = 10 + 0.00998
= **10.0 mol dm^{-3}**

You may find the use of ICE table helpful.

	$HSO_4^-(aq)$	$\rightleftharpoons$	$H^+(aq)$	+ $SO_4^{2-}(aq)$
initial conc. / mol dm^{-3}	10		10	–
change in conc. / mol dm^{-3}	– *x*		+ *x*	+ *x*
conc. at eqm / mol dm^{-3}	10 – x		10 + x	x

(vi) $[SO_4^{2-}]$ = **0.0100 mol dm^{-3}** (accept 0.00998 mol dm^{-3}) [1]

$[HSO_4^-]$ = 10 – 0.01
= **9.99 mol dm^{-3}**

(vii) HSO_4^- is a very weak base because its tendency to accept a H^+ proton is very low since H_2SO_4 is completely ionised / a strong acid. [1]

or K_b (HSO_4^-) = $\frac{K_w}{K_a(H_2SO_4)}$ is very small as H_2SO_4 is a strong acid and its K_a is large.

Hence HSO_4^- is a stronger acid than it is a base as K_a (HSO_4^-) = 1.0 x 10^{-2} >> K_b (HSO_4^-). [1]

(b) **(i)** Anode: **$2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e$ or $2H_2SO_4 \longrightarrow H_2S_2O_8 + 2H^+ + 2e$** [2]

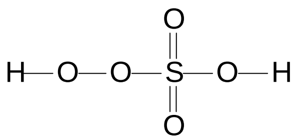
Cathode: **$2H^+ + 2e \longrightarrow H_2$**
(do not accept $2H_2O + 2e \longrightarrow H_2 + 2OH^-$)

(ii) **$H_2S_2O_8 + 2H_2O \longrightarrow 2H_2SO_4 + H_2O_2$** [1]

(iii) **$H_2S_2O_8 + H_2O \longrightarrow H_2SO_4 + H_2SO_5$** [2]
$H_2SO_5 + H_2O \longrightarrow H_2SO_4 + H_2O_2$

[1]: each correct step

(iv) [3]



$\angle$HOO = 104.5^o (accept 104^o – 105^o)
$\angle$OOS = 104.5^o (accept 104^o – 105^o)
$\angle$SOH = 104.5^o (accept 104^o – 105^o)
$\angle$OSO = 109.5^o (accept 109^o)

[1]: correct displayed structure
[1]: each correct bond angles wrt O and S (ignore repeated bond angle)

(v) O.N. of S in $H_2S_2O_8$ and H_2SO_5 = **+6** [1]

(c) decomposition of H_2O_2: $2H_2O_2 \longrightarrow 2H_2O + O_2$ [2]

$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$	$E^\ominus = +1.77$ V
$O_2 + 2H^+ + 2e^- \rightleftharpoons H_2O_2$	$E^\ominus = +0.68$ V
$Mn^{3+} + e^- \rightleftharpoons Mn^{2+}$	$E^\ominus = +1.49$ V

(Step 1: $2Mn^{2+} + H_2O_2 + 2H^+ \longrightarrow 2Mn^{3+} + 2H_2O$) $E^\ominus_{cell}$ = **+0.28V**
(Step 2: $2Mn^{3+} + H_2O_2 \longrightarrow 2Mn^{2+} + O_2 + 2H^+$) $E^\ominus_{cell}$ = **+0.81V**

Since $E^\ominus_{cell} > 0$, both reactions are feasible. Hence Mn^{2+} acts as a homogeneous catalyst in the two–step reaction by providing an alternative pathway with lower activation energy via the intermediate Mn^{3+}(aq).

[1]: suitable choice of $E^\ominus$ wrt Mn^{2+} (accept MnO_4^-/Mn^{2+} but not MnO_2/Mn^{2+})
[1]: appropriate attempts to show $E^\ominus_{cell} > 0$ (i.e. ignore numerical error if method and conclusion is correct)
(award only the 2nd mark if Mn^{3+} is used as catalyst instead)

3 **(a)** **(i)** high temperature and low pressure [1]

(ii) [2]

$$n(PH_3) = \frac{68}{[31.0 + 3(1.0)]} = 2.00 \text{ mol}$$

$$pV = nRT$$

$$V = \frac{nRT}{p}$$

$$= \frac{2.00(8.31)(800)}{3.36 \times 10^7}$$

$$= 3.96 \times 10^{-4} \text{ m}^3$$

$$= \mathbf{396 \text{ cm}^3}$$

[1]: correct volume in m^3
[1]: correct conversion from m^3 to cm^3

(iii) PH_3 behaves non–ideally and volume calculated in **(a)(ii)** is smaller than the volume of the container. [2]

under high pressure,
the volume of the PH_3 molecules is not negligible / significant. Hence the actual free volume between the gas molecules is smaller.

[1]: recognise that PH_3 is behaving non–ideally and volume is smaller
[1]: correct explanation

(b) **(i)** [1]

$$\frac{1}{4}P_4(s) + \frac{3}{2}H_2(g) \longrightarrow PH_3(g)$$

(with correct state symbols, accept if just use P)

(ii) By Hess's Law, [2]

$$\Delta H_f^{\ominus} = \frac{1}{4}(-3012) + \frac{3}{2}(-286) - (-1291)$$

$$= \mathbf{+109 \text{ kJ mol}^{-1}}$$

[1]: correct application of Hess' Law
[1]: correct answers with units

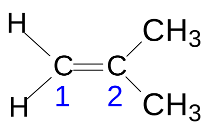
(iii) C*l* is more electronegative than H. Hence the H–C*l* bond is polar and the bond has ionic character which results in extra bond strength. [1]

The electronegativity of P and H are similar, hence the bond energy of the P–H bond is just the average of the H–H and P–H bond energies.

(c) **(i)** nucleophilic substitution [1]

(ii) It acts as a base. [1]

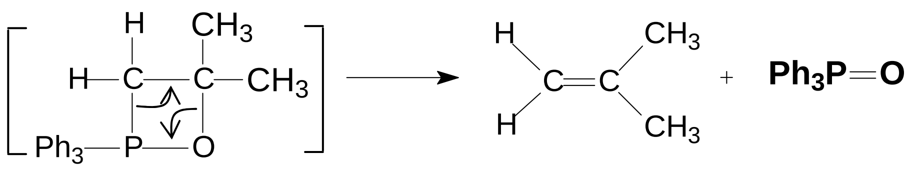
(iii) [2]



oxidation number of
C1: **–2**
C2: **0**

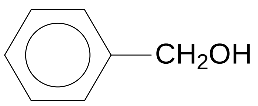
[1]: each correct answer

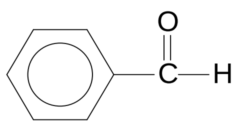
(iv) [2]



[1]: both curly arrows correctly shown (allow 4)
[1]: correct by–product

(d) Step V: $NaOH(aq)$, heat (under reflux) [5]
Step VI: $K_2Cr_2O_7$, dilute H_2SO_4, warm with (immediate) distillation

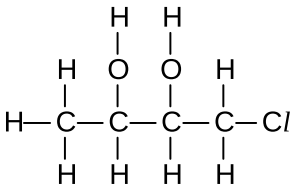
K: 

L: 

M: $Ph_3\overset{+}{P}–\overset{-}{C}HPh$

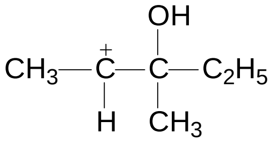
[1] each

4 (a) (i) [2]



[1]: correct diol
[1]: correct displayed formula shown

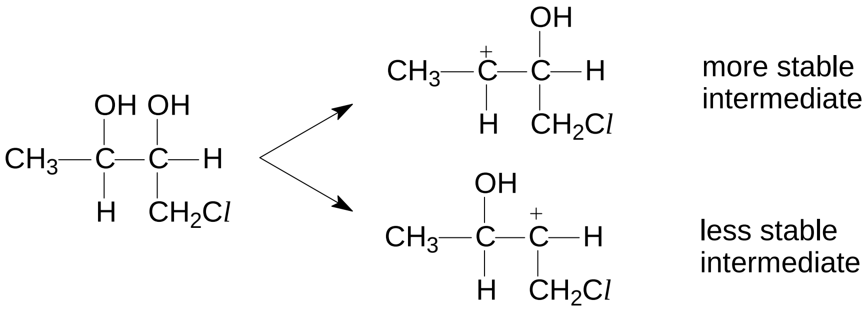
(ii) [2]



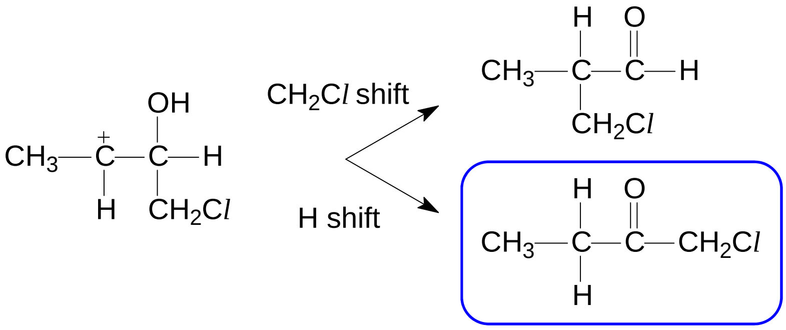
The yield of carbocation **Q** is smaller as it is a less stable carbocation due to 1 less electron donating alkyl group to help disperse the positive charge on the carbocation.

[1]: correct structure of **Q**
[1]: correct explanation

(iii) [1]

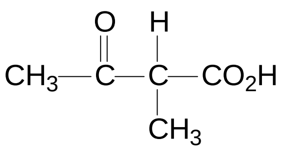


followed by

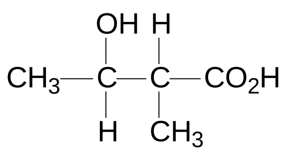


[1]: correct carbonyl compound

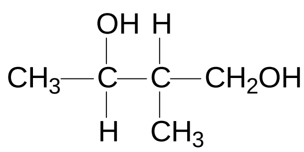
(b) (i) [3]



R



S

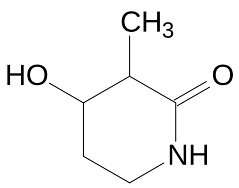


T

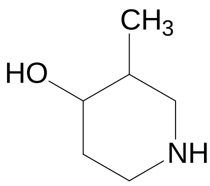
[1] each

Not required
R and **S** react with Na_2CO_3.
⇒ presence of $-CO_2H$
⇒ likely presence of 1° in **T** (due to reduction of $-CO_2H$ in **R**)
R, **S** and **T** give yellow ppt with I_2/OH^-.
⇒ presence of CH_3COR or in **R** and $CH_3CH(OH)R$ in **S** and **T** (**S** and **T** are formed from the reduction of **R**)
From these 2 pieces of information, you can deduce the structure of **R** as the remaining 2 C atoms must be branched in order for **R** to be chiral.

(ii) [2]



V



W

[1] each

Not required
V and **W** react with sodium metal.
⇒ likely presence of (2°) alcohol (due to reduction of ketone in **U**)
Only **W** reacts with HCl(aq).
⇒ presence of basic amine group in **W** (infer from the molecular formula and structure of **U**)
From these 2 pieces of information and from the loss of 1 O atom and gain of 4 H atoms (**U** ⟶ **W**), you can deduce the structure of **W**. (or by comparing **W** with **V**)
Other cues include "$LiAlH_4$ is the more powerful of the two" and "behaves differently towards the two reducing agents".

(iii) Add 2,4–dintrophenylhydrazine to separate test–tubes containing **U** and **V**. [2]
U gives an orange ppt but no orange ppt observed for **V**.

(accept use of anhydrous PCl_5 / $SOCl_2$ or $K_2Cr_2O_7$, dil H_2SO_4)

[1]: correct test
[1]: correct observation

(iv) Compound **U** is neutral but ethylamine is basic. Lone pair on the N of **U** is delocalised into the C=O, hence it is not available to accept the H^+ ion. [2]
In ethylamine, the electron donating ethyl group bonded to N makes the lone pair more available to accept the H^+ ion, hence it is more basic.

[1]: correct relative basicity
[1]: correct explanation

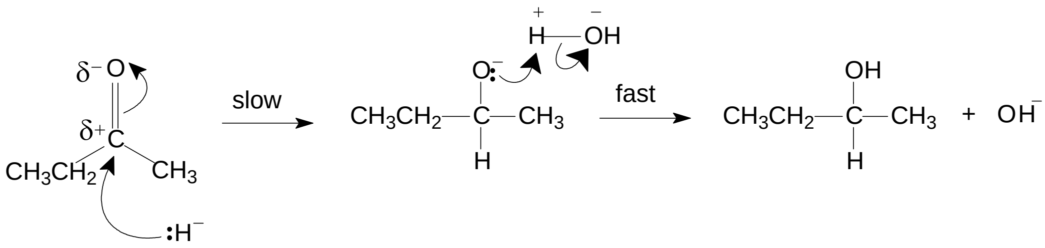
(c) $\mathbf{LiAlH_4 + 4H_2O \longrightarrow LiOH + Al(OH)_3 + 4H_2}$ [1]

$\mathbf{LiOH + Al(OH)_3 + 4H^+ \longrightarrow Li^+ + Al^{3+} + 4H_2O}$ [1]

(d) **(i)** nucleophilic addition [2]

(Generation of nucleophile: $LiAlH_4 \longrightarrow LiAlH_3^+ + H^-$)

δ δ



[1]: correct dipoles on the carbonyl C and O and correct curly arrows
[1]: correct intermediate and product, (slow/fast steps clearly indicated)

(ii) Since the butanone molecule is trigonal planar with respect to the carbonyl carbon atom, the nucleophile has equal probability of attacking it from either side of the molecule. [1]
This generates equimolar mixture of the enantiomers / alcohol formed via nucleophilic addition mechanism and gives an optically inactive racemic mixture which does not rotate plane–polarised light.

(iii) Absence of electron deficient C atom / polarised bond or same electronegativity between 2 C atoms and not susceptible to nucleophilic attack by H^-. [1]

5 (a) (i) [1]

$$D = \frac{[I_2]_{CCl_4}}{[I_2]_{water}}$$

$$= \frac{0.102}{0.00120} = \mathbf{85}$$

OR

$$D = \frac{[I_2]_{water}}{[I_2]_{CCl_4}} = \frac{0.00120}{0.102} = \mathbf{0.0118}$$

(ii) When the $[I^-]$ increases, by Le Chatelier's Principle, the <u>position of equilibrium shifts to the right</u>, to <u>decrease the concentration of I^-</u>, hence favouring the forward reaction. This results in a decrease in the fraction of I_2. [1]

(iii) [1]

$$85 = \frac{[I_2]_{CCl_4}}{[I_2]_{water}} = \frac{0.085}{[I_2]_{water}}$$

$$[I_2]_{water} = \frac{0.085}{85}$$

$$= \mathbf{0.001\ mol\ dm^{-3}}$$

(iv) Since $[I_2]_{total} = [I_2(aq)] + [I_3^-(aq)]$ [2]

$0.01 = 0.001 + [I_3^-(aq)]$

$[I_3^-(aq)] =$ **$0.009\ mol\ dm^{-3}$**

	$I_2(aq)$	+ $I^-(aq)$	$\rightleftharpoons$	$I_3^-(aq)$
initial conc. / mol dm^{-3}		0.0232		–
change in conc. / mol dm^{-3}		*– 0.009*		*+ 0.009*
conc. at eqm / mol dm^{-3}	0.001	**0.0142**		0.009

[1]: correct $[I_3^-(aq)]$
[1]: correct $[I^-(aq)]$

(vi) [2]

$$K_c = \frac{0.009}{(0.001)(0.0142)}$$

$$= \mathbf{634\ mol^{-1}\ dm^{3}}$$

[1]: correct answer
[1]: correct units

(b) (i) **A** = $Co(OH)_2(H_2O)_4$ (accept $Co(OH)_2$) [1]

(ii) [2]

$[Co(NH_3)_6]^{3+} + e^- \rightleftharpoons [Co(NH_3)_6]^{2+}$ $E^\ominus = +0.11V$
$O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-$ $E^\ominus = +0.40V$

$E^\ominus_{cell} = +0.40 - (+0.11)$
$= \mathbf{+0.29V} > 0$

Hence yellow–brown solution **B** changes to red–brown solution **C** as the $[Co(NH_3)_6]^{2+}$ is oxidised to $[Co(NH_3)_6]^{3+}$.

[1]: correct choice of $E^\ominus$ ($O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O$ not accepted)
[1]: correct $E^\ominus_{cell}$ calculated (allow ecf)

(iii) When concentrated HC*l* is added to an aqueous solution of Co^{2+}, ligand exchange occurs where Cl^- displaces H_2O ligands, forming the blue $[CoCl_4]^{2-}$. [2]

$$[Co(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CoCl_4]^{2-} + 6H_2O$$

Addition of water will cause the position of equilibrium to shift to the left, reforming the pink $[Co(H_2O)_6]^{2+}$.

[1]: correct equation (accept $[CoCl_6]^{4-}$)
[1]: correct explanation

Note: Addition of water will result in decrease in concentration of the ions. Position of equilibrium will shift to the left as it will result in an increase in no. of ions, reforming the pink $[Co(H_2O)_6]^{2+}$ as shown by the equation
$[CoCl_4]^{2-} + 6H_2O \rightleftharpoons [Co(H_2O)_6]^{2+} + 4Cl^-$

(recall: the effect of decreasing pressure on system will cause P.O.E. to shift to the side with more gaseous particles so as to increase the pressure)

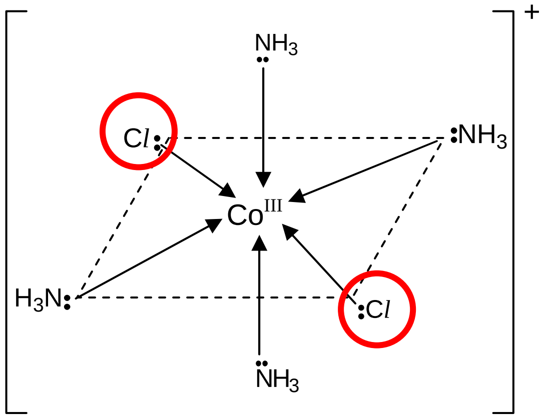
(c) **(i)** In the presence of ligands, [3]

- the partially–filled 3d orbitals of Co^{3+} are split into two levels (non–degenerate) with a small energy gap E
- When light passes through, photons/energy equal to red light is absorbed.
- An electron promotes from a lower energy d orbital to a vacant/unfilled/partially filled higher energy d orbital, i.e. d–d transition,
- The green colour of the complex **F** is complementary to red absorbed.

[1]: partially filled 3d–orbital; small energy gap
[1]: d–d transition
[1]: absorption of red (complementary to green)

(ii) Number of moles of free Cl^- per mole of complex [2]
= 2 x number of moles of $PbCl_2$ precipitated per mole of complex

F: 0.5 mol of $PbCl_2$ precipitated
$\Rightarrow$ 1 mol of free Cl^- and 2 mol of Cl^- as ligand per mol of complex
$\Rightarrow$ cation is $\mathbf{[Co(NH_3)_4Cl_2]^+}$
$\Rightarrow$ the 2 Cl^- ligands must be at 180° to each other / cannot be 90° away from each other (no net dipole)



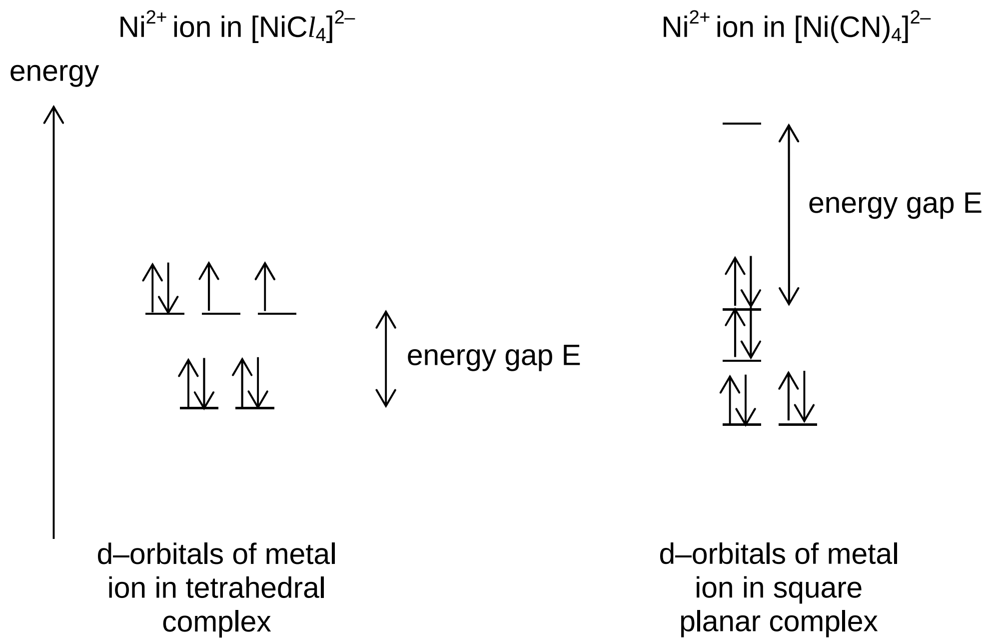
F

[1]: correct formula for complex ion
[1]: for structure with lone pairs and correct net charge

(d) **(i)** The energy gap E between the two sets of d–orbitals is greater than the repulsion energy, favouring the pairing of electrons. [1]

[1]: for comparison of energy gap E and repulsion energy between electrons

(ii) Ni^{2+} ion: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8$ [2]



In $[NiCl_4]^{2-}$, there are 2 unpaired electrons in the d–orbitals. Hence it is paramagnetic.
In $[Ni(CN)_4]^{2-}$, there are 0 unpaired electrons in the d–orbitals. Hence it is non–paramagnetic.

[1]: correct electronic configuration / no. of d–orbital electrons for Ni^{2+}
[1]: correct placement of electrons